

05

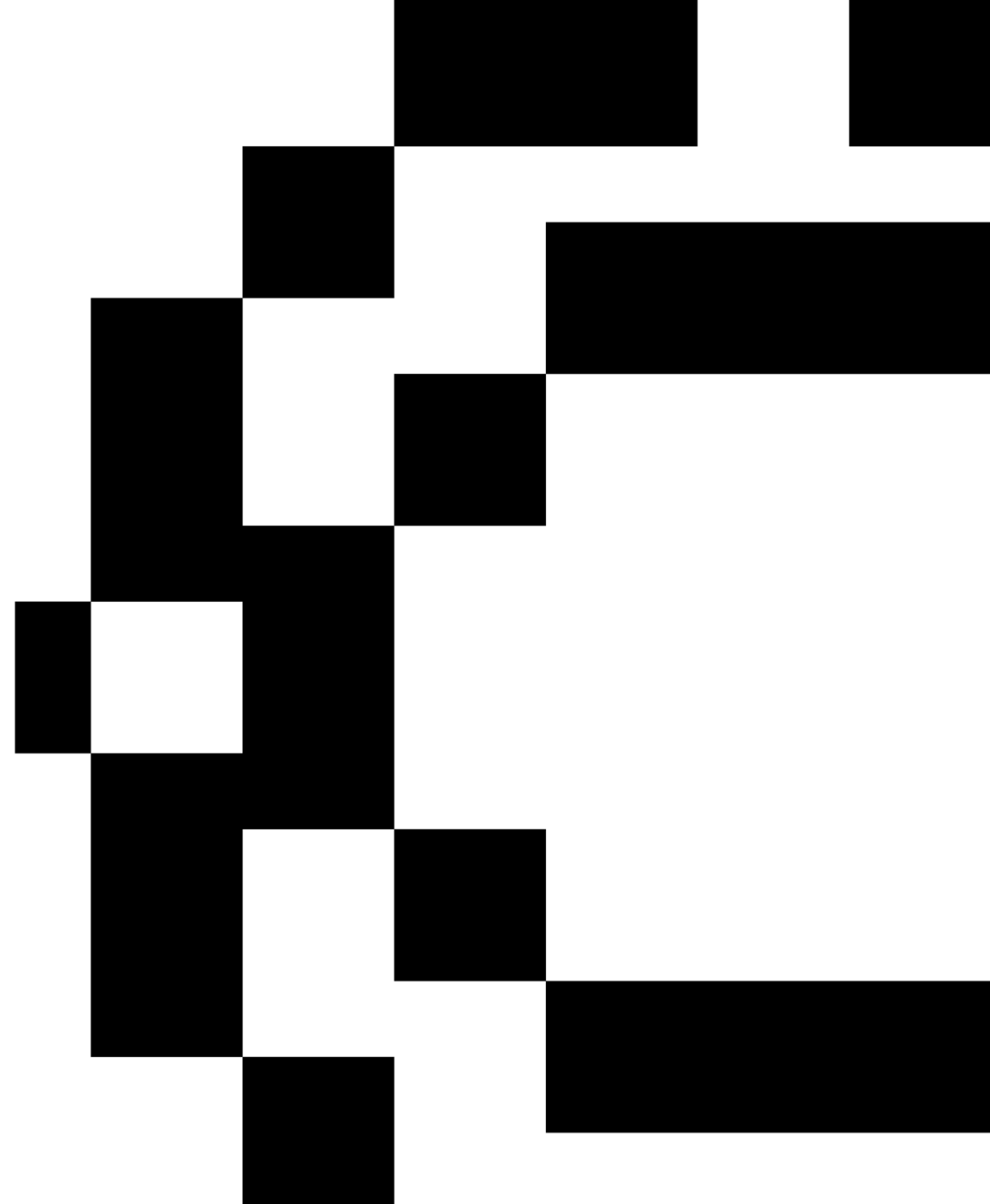
Feature Selection

Data Mining II| MGI 2526

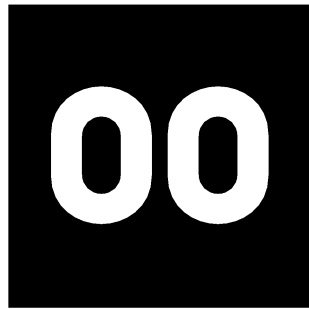
March 16th and 17th

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Agenda



**Feature
Selection**



**Filter
Methods**



**Wrapper
Methods**



**Embedded
Methods**

00. Feature Selection

Process to **select automatically or manually a subset of relevant features** to use in model building

0.1. Feature Selection

The Importance of Feature Selection



The goal

Reduce the number of input variables to those that are believed to be most useful to a model in order to predict the target variable



Why should we care?

- **Performance improvement:** Enhanced generalization by reducing overfitting
- **Computational cost decrease:** Shorter training time
- Simple models are **easier to understand**
- Variable **redundancy**

Specific cases:

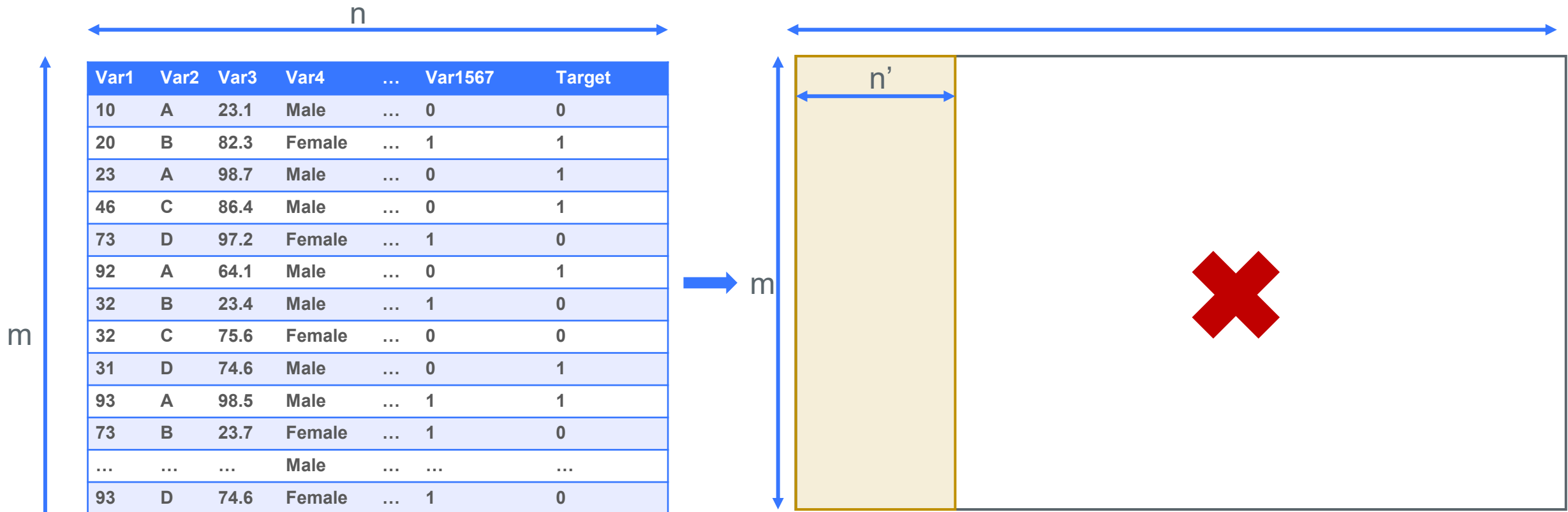
- SVMs and Neural Networks are vulnerable to irrelevant predictors
 - Not relevant predictors can shrink the predictive performance of the models
- Linear and Logistic Regressions are sensible to correlated predictors
 - Removing redundancy will reduce multicollinearity and allow a better fit of the models

0.1. Feature Selection

The Importance of Feature Selection

Thousands to millions of low-level features

- Select the most relevant ones to build better, faster and easier to understand models



0.2. Feature Selection

Feature Selection vs Feature Engineering vs Dimensionality Reduction

Feature selection vs Feature Engineering

Feature engineering aims to create new features from the original ones with the goal of creating models more effective and with higher performance

Feature selection allow to select features from all the features available (the original ones and the created ones) to allow the creation of more efficient models

In a “normal” pipeline, we perform feature engineering before applying feature selection

0.2. Feature Selection

Feature Selection vs Feature Engineering vs Dimensionality Reduction

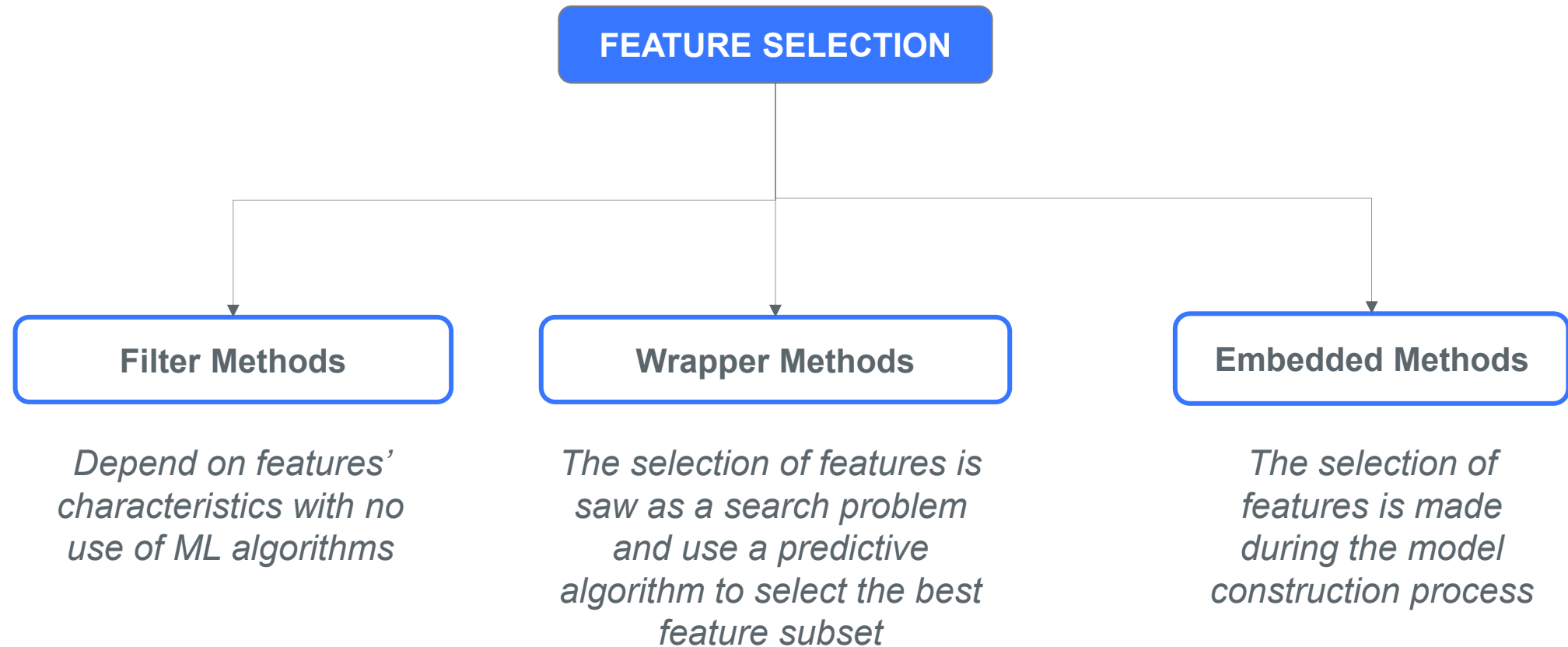
Feature selection vs Dimensionality reduction

Dimensionality reduction use unsupervised algorithms to reduce the number of features in a dataset. Those techniques modify or transform features into a lower dimension.

Feature selection also reduce the number of features in a dataset, however, is a process that select and exclude features without any kind of modification.

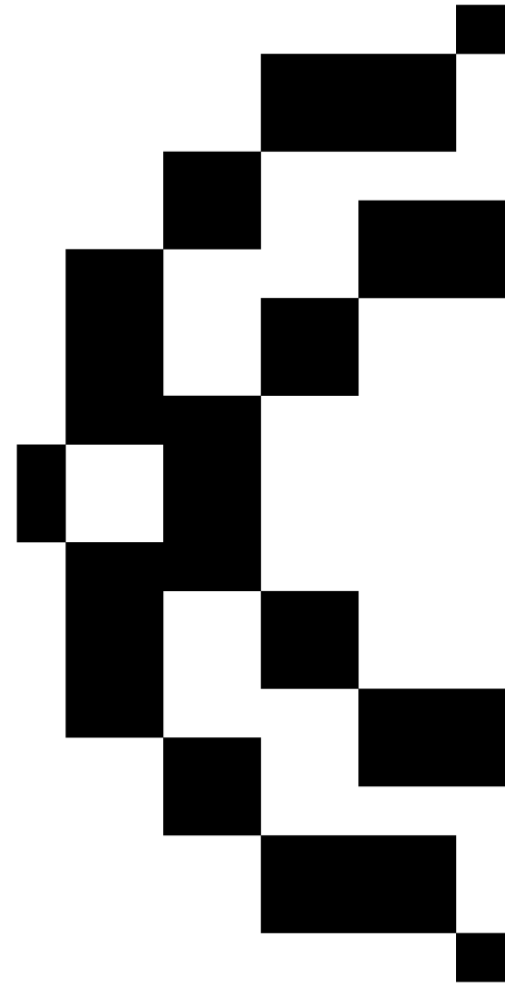
0.3. Feature Selection

Feature Selection Methods



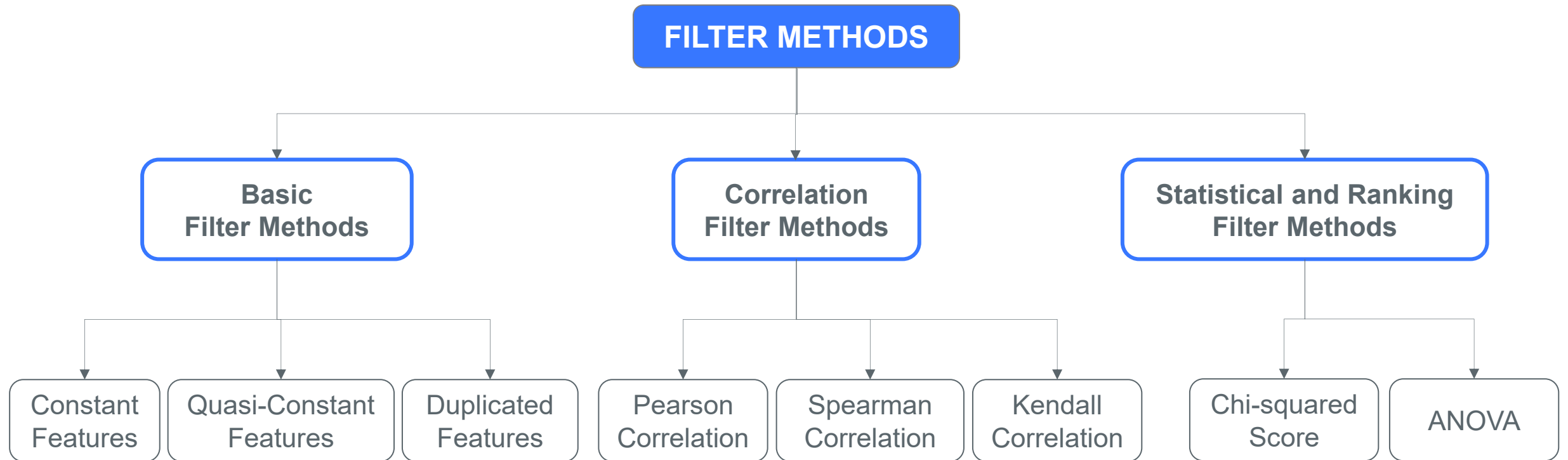
00. Filter Methods

Filter methods are techniques used to select features from a dataset **without the use of any machine learning algorithm**



1. Filter Methods

Introduction



1.1. Filter Methods

Basic Filter Methods

ID	Variable_1	Variable_2	Variable_3	Variable_4	Target
1001	Female	Yes	Portugal	PT	0
1002	Female	Yes	Spain	SP	1
1003	Female	Yes	France	FR	1
1004	Female	Yes	Portugal	PT	1
1005	Female	Yes	Portugal	PT	0
1006	Female	Yes	United Kingdom	UK	1
1007	Female	No	Spain	SP	0
1008	Female	Yes	Portugal	PT	0
1009	Female	Yes	France	FR	1
1010	Female	Yes	United Kingdom	UK	1
1011	Female	Yes	Portugal	PT	0
...	...	...	...	...	...
10000	Female	Yes	France	FR	0

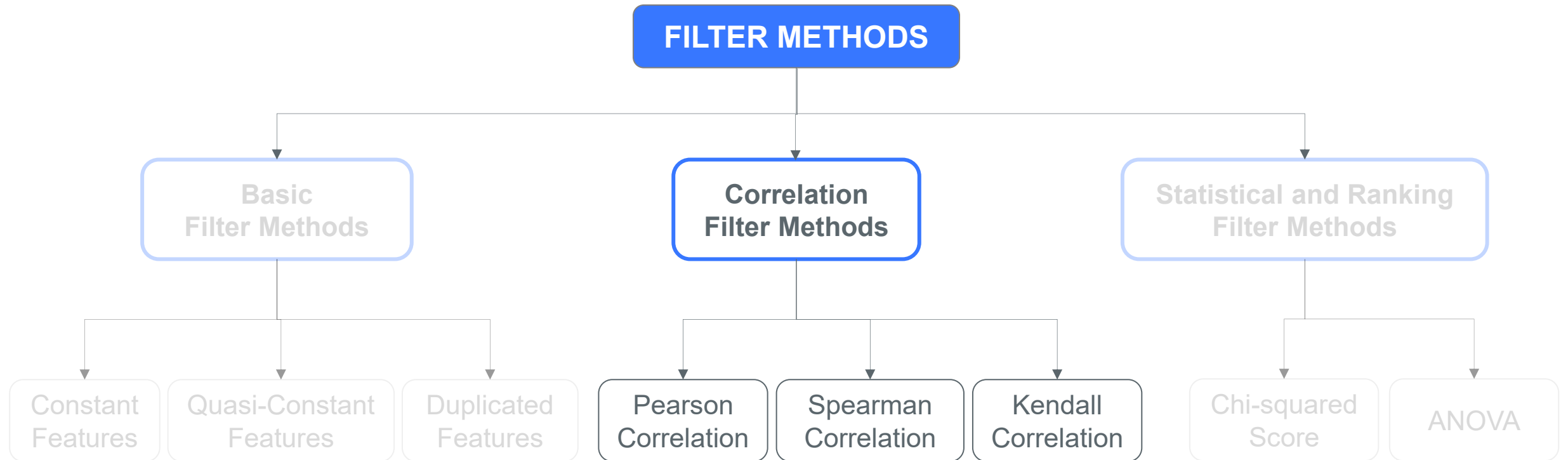
Constant
features

Quasi-
constant
features

Duplicate
Features

1.2. Filter Methods

Correlation Filter Methods



1.2. Filter Methods

Correlation Filter Methods

Correlation measures the degree to which two variables move in relation to each other.

Are correlated variables always bad?

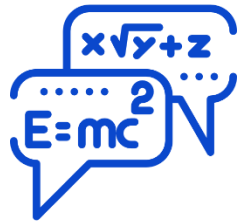
- If two predictors are highly correlated among themselves, they provide redundant information. We can make an accurate prediction on the target with just one of those redundant variables.
- If a predictor is highly correlated with the target, this is a useful property and should be used in the final subset of variables chosen

If variable A and variable B are highly correlated among themselves, and we should remove one of those, should we keep the variable A or B?

- Use a feature selection technique to understand the weight of each variable in the target. Keep the one that is the most important!

1.2.1. Filter Methods

Correlation Filter Methods – Pearson Correlation



The formula

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{(x - \bar{x})^2} \sqrt{(y - \bar{y})^2}}$$

The mean of the variable x

The mean of the variable y

INPUT VARIABLE: Continuous
OUTPUT VARIABLE: Continuous

1.2.1. Filter Methods

Correlation Filter Methods – Pearson Correlation



Example

ID	x	y	$(x - \bar{x})$	$(y - \bar{y})$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
1001	87	241	10	27	270	100	729
1002	46	160	-31	-54	1674	961	2916
1003	72	210	-5	-4	20	25	16
1004	64	195	-13	-19	247	169	361
1005	123	285	46	-71	3266	2116	5041
1006	97	233	20	19	380	400	361
1007	50	174	-27	-40	1080	729	1600
Mean	77	214	-	-			
Sum	-	-	-	-	6937	4500	11024
$\sqrt{\text{Sum}}$						67.08	105

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{(x - \bar{x})^2} \sqrt{(y - \bar{y})^2}} = \frac{6937}{67.08 \times 105} = 0.985$$

1.2.1. Filter Methods

Correlation Filter Methods – Pearson Correlation



The range

- Can vary between -1 and 1
- 1 means a positive correlation: the values of one variable increase as the values of another increase
- -1 means a negative correlation: the values of one variable decrease as the values of another increase
- 0 means no linear correlation between the two variables

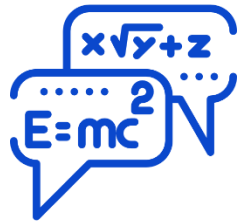


Assumptions

- Both variables should be normally distributed
- There is a linear relationship between the two variables

1.2.2. Filter Methods

Correlation Filter Methods – Spearman Correlation



The formula

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Difference between two rankings

Number of observations

INPUT VARIABLE: Continuous or ordinal
OUTPUT VARIABLE: Continuous or ordinal

1.2.2. Filter Methods

Correlation Filter Methods – Spearman Correlation



Example

ID	x	y	Rank x	Rank y	d_i^2
1001	87	241	5	6	1
1002	46	160	1	1	0
1003	72	210	4	4	0
1004	64	195	3	3	0
1005	123	285	7	7	0
1006	97	233	6	5	1
1007	50	174	2	2	0
Sum	-	-			2

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 2}{7(49 - 1)} = 0.964$$

1.2.2. Filter Methods

Correlation Filter Methods – Spearman Correlation



The range

- Can vary between -1 and 1
- 1 means a positive correlation: the values of one variable increase as the values of another increase
- -1 means a negative correlation: the values of one variable decrease as the values of another increase
- 0 means no correlation between the two variables

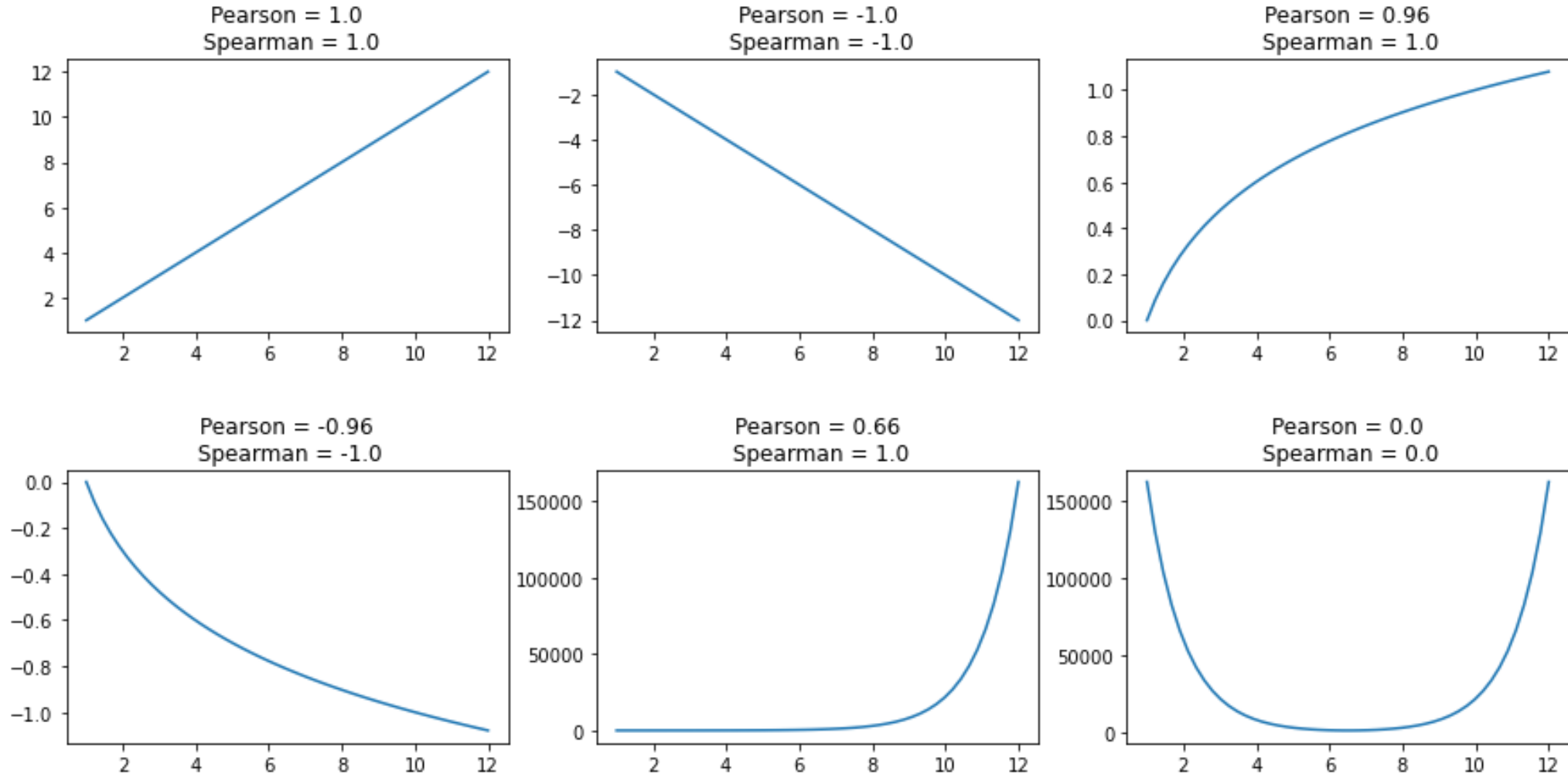


Assumptions

- There are no assumptions on the distribution of the variables

1.2. Filter Methods

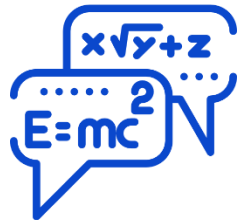
Correlation Filter Methods – Spearman Correlation vs Pearson Correlation



1.2.3. Filter Methods

Correlation Filter Methods – Kendall Correlation

Kendall correlation is a coefficient that represents the degree of concordance between two ranked variables.



The formula

$$\tau = \frac{2(n_c - n_d)}{n(n-1)}$$

Number of concordant pairs

Number of discordant pairs

Number of observations

INPUT VARIABLE: Continuous or ordinal
OUTPUT VARIABLE: Continuous or ordinal

1.2.3. Filter Methods

Correlation Filter Methods – Kendall Correlation

- It calculates a normalized score for the number of matching / concordant rankings between the two variables.
 - The pair is concordant if $X_i > X_j$ and $Y_i > Y_j$ or $X_i < X_j$ and $Y_i < Y_j$
 - The pair is discordant if $X_i > X_j$ and $Y_i < Y_j$ or $X_i < X_j$ and $Y_i > Y_j$

ID	x	y
1001	87	241
1002	46	160
1003	72	270



$X_{1001} > X_{1002}$ and $Y_{1001} > Y_{1002} \rightarrow \text{concordant pair}$

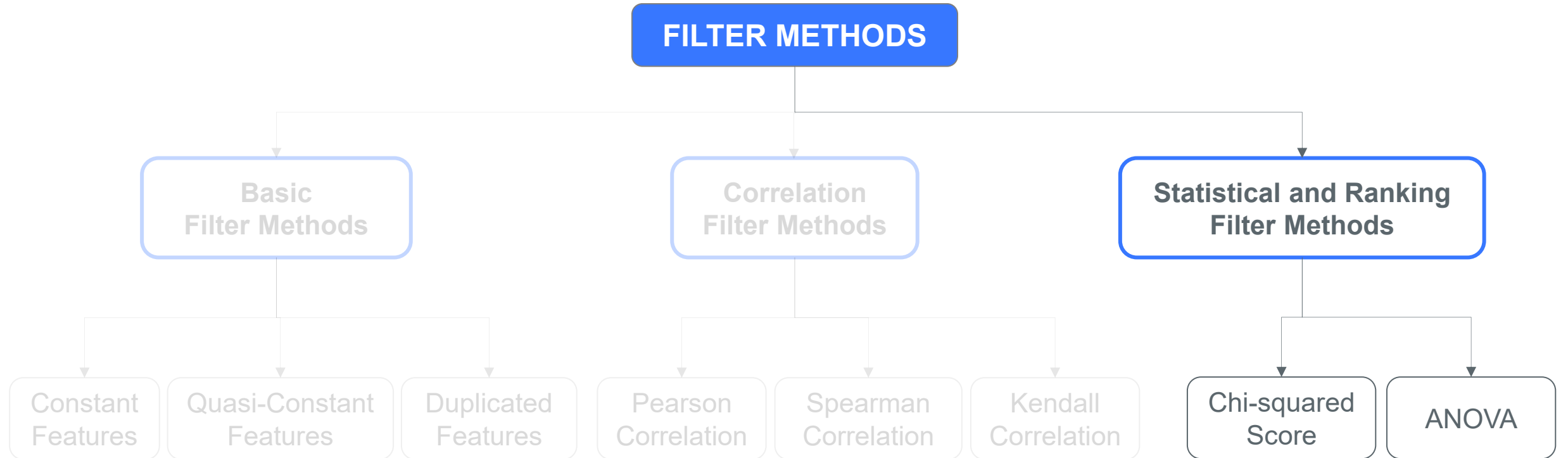
$X_{1001} > X_{1003}$ and $Y_{1001} < Y_{1003} \rightarrow \text{discordant pair}$

$X_{1002} < X_{1003}$ and $Y_{1002} < Y_{1003} \rightarrow \text{concordant pair}$

- Kendall's correlation varies between -1 and 1:
 - 1 (High) – Observations have a similar rank between the two variables
 - -1 (Low) – Observations have a dissimilar rank between the two variables

1.3. Filter Methods

Statistical and Ranking Filter Methods



1.3.1. Filter Methods

Statistical and Ranking Filter Methods – Chi-Square Score

Statistical method used to test the independence of two events

- We want to determine the relationship between a predictor and the target
- In feature selection, the goal is to identify and select the features which are highly dependent on the response / target
- Given two variables, we can get observed count O and expected count E
 - When two features are independent, the observed count is close to the expected count, thus we will have smaller Chi-Square value.
 - The higher the Chi-square value implies, in simple words, that the feature is more dependent on the target, and it can be selected for model training.

INPUT VARIABLE: Categorical
OUTPUT VARIABLE: Categorical

1.3.1. Filter Methods

Statistical and Ranking Filter Methods – Chi-Square Score

$$\chi^2 = \sum \frac{(\text{Observed value} - \text{Expected value})^2}{\text{Expected value}}$$

Step 1 – Define hypothesis

Null hypothesis (H0) : Two variables are independent

Alternate hypothesis (H1): Two variables are not independent

1.3.1. Filter Methods

Statistical and Ranking Filter Methods – Chi-Square Score

Step 2 – Define the contingency table for observed values

Gender	Disease
Female	Yes
Male	No
Female	Yes
Female	No
Male	No
Female	No
Male	Yes
Female	Yes
Male	No
...	
Male	Yes



Gender	Yes	No	Total
Male	380	1780	2160
Female	440	1400	1840
Total	820	3180	4000

Contingency table for observed values

Show the distribution of one variable in rows and another in columns.

Used to study the relation between two variables

Degrees of freedom for contingency table: given as $(r - 1) \times (c - 1)$, where r and c are rows and columns respectively.

In the example: $df = (2 - 1) \times (2 - 1) = 1$

1.3.1. Filter Methods

Statistical and Ranking Filter Methods – Chi-Square Score

Gender	Yes	No	Total
Male	380	1780	2160
Female	440	1400	1840
Total	820	3180	4000

Step 3 – Find the expected value

$$P(A \cap B) = P(A) \times P(B)$$

$$P(Yes \cap Male) = P(Yes) \times P(Male) = \left(\frac{820}{4000}\right) \times \left(\frac{2160}{4000}\right) = 0.1107 \rightarrow 4000 \times 0.1107 = 443$$

$$P(Yes \cap Female) = P(Yes) \times P(Female) = \left(\frac{820}{4000}\right) \times \left(\frac{1840}{4000}\right) = 0.0943 \rightarrow 4000 \times 0.0943 = 377$$

$$P(No \cap Male) = P(No) \times P(Male) = \left(\frac{3180}{4000}\right) \times \left(\frac{2160}{4000}\right) = 0.4293 \rightarrow 4000 \times 0.4293 = 1717$$

$$P(No \cap Female) = P(No) \times P(Female) = \left(\frac{3180}{4000}\right) \times \left(\frac{1840}{4000}\right) = 0.3657 \rightarrow 4000 \times 0.3657 = 1463$$

Gender	Yes	No
Male	443	1717
Female	377	1463

Expected Values

1.3.1. Filter Methods

Statistical and Ranking Filter Methods – Chi-Square Score

Step 4 – Calculate chi-square value

$$\chi^2 = \sum \frac{(\text{Observed value} - \text{Expected value})^2}{\text{Expected value}}$$

Observed values

Gender	Yes	No	Total
Male	380	1780	2160
Female	440	1400	1840
Total	820	3180	4000



Expected values

Gender	Yes	No
Male	443	1717
Female	377	1463

Gender, Disease	<i>O</i>	<i>E</i>	<i>(O - E)</i>	<i>(O - E) ^ 2</i>	<i>(O - E) ^ 2 / E</i>
Male, Yes	380	443	-63	3969	8.96
Male, No	1780	1717	63	3969	2.31
Female, Yes	440	377	63	3969	10.53
Female, No	1400	1463	-63	3969	2.71
Chi-Square Value					24.51

Null hypothesis (H0) : Two variables are independent
 Alternate hypothesis (H1): Two variables are not independent

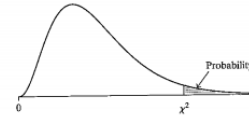
1.3.1. Filter Methods

Statistical and Ranking Filter Methods – Chi-Square Score

Step 5 – Accept or reject the null hypothesis

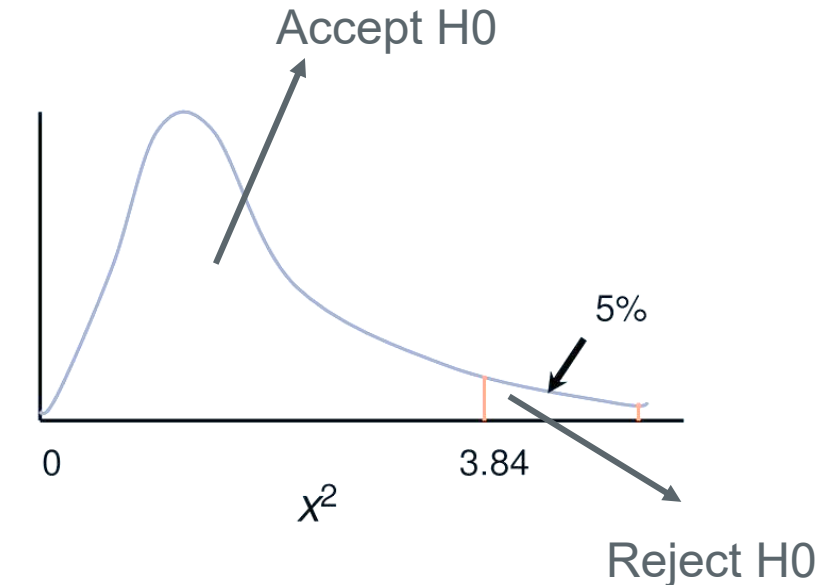
Chi-Square table

TABLE C: Chi-Squared Distribution Values for Various Right-Tail Probabilities



df	Right-Tail Probability						
	0.250	0.100	0.050	0.025	0.010	0.005	0.001
1	1.32	2.71	3.84	5.02	6.63	7.88	10.83
2	2.77	4.61	5.99	7.38	9.21	10.60	13.82
3	4.11	6.25	7.81	9.35	11.34	12.84	16.27
4	5.39	7.78	9.49	11.14	13.28	14.86	18.47
5	6.63	9.24	11.07	12.83	15.09	16.75	20.52
6	7.88	10.64	12.59	14.45	16.81	18.55	22.46
7	9.04	12.02	14.07	16.01	18.48	20.28	24.32
8	10.22	13.36	15.51	17.53	20.09	21.96	26.12
9	11.39	14.68	16.92	19.02	21.67	23.59	27.88
10	12.55	15.99	18.31	20.48	23.21	25.19	29.59
11	13.70	17.28	19.68	21.92	24.72	26.76	31.26
12	14.85	18.55	21.03	23.34	26.22	28.30	32.91
13	15.98	19.81	22.36	24.74	27.69	29.82	34.53
14	17.12	21.06	23.68	26.12	29.14	31.32	36.12
15	18.25	22.31	25.00	27.49	30.58	32.80	37.70
16	19.37	23.54	26.30	28.85	32.00	34.27	39.25
17	20.49	24.77	27.59	30.19	33.41	35.72	40.79
18	21.60	25.99	28.87	31.53	34.81	37.16	42.31
19	22.72	27.20	30.14	32.85	36.19	38.58	43.82
20	23.83	28.41	31.41	34.17	37.57	40.00	45.32
25	29.34	34.38	37.65	40.65	44.31	46.93	52.62
30	34.80	40.26	43.77	46.98	50.89	53.67	59.70
40	45.62	51.80	55.76	59.34	63.69	66.77	73.40
50	56.33	63.17	67.50	71.42	76.15	79.49	86.66
60	66.98	74.40	79.08	83.30	88.38	91.95	99.61
70	77.58	85.53	90.53	95.02	100.4	104.2	112.3
80	88.13	96.58	101.88	106.6	112.3	116.3	124.8
90	98.65	107.6	113.1	118.1	124.1	128.3	137.2
100	109.1	118.5	124.3	129.6	135.8	140.2	149.5

With 95% of confidence (alpha = 0.05)
 With degrees of freedom = 1



Chi-square value = 3.84

Conclusion: We reject the null hypothesis since the Chi-Square value is higher than the critical chi-square value. The two variables are not independent, and Gender should be selected for training the model.

1.3.2. Filter Methods

Statistical and Ranking Filter Methods – ANOVA

- Statistical method used to check the means of two or more groups that are significantly different from each other
- ANOVA assumes a linear relationship between the variables and the target
- ANOVA assumes that the variables are normally distributed

INPUT VARIABLE: Categorical
OUTPUT VARIABLE: Continuous

1.2. Filter Methods

Advantages

- Selected features can be used in any machine learning algorithm
- Computationally inexpensive

1.2. Filter Methods

When to Use them? (RECAP)

Pearson correlation

- Input variable continuous, Output variable continuous

Spearman correlation

- Input variable continuous or ordinal, Output variable continuous or ordinal

Kendall correlation

- Input variable continuous or ordinal, Output variable continuous or ordinal

Chi-Square

- Input variable categorical, Output variable categorical

One-way ANOVA

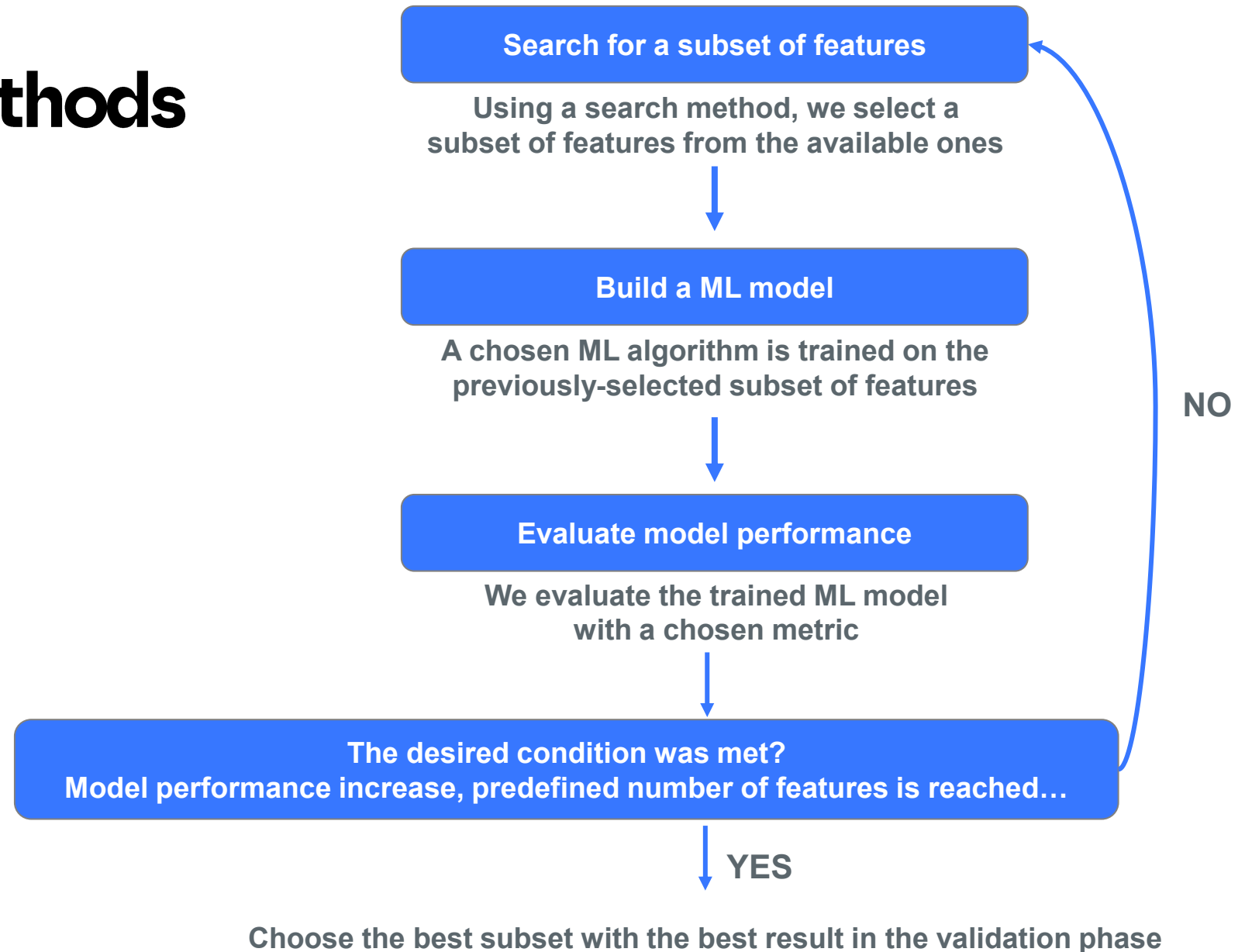
- Input Variable Categorical, Output Variable continuous

02. Wrapper Methods

Wrapper methods **use machine learning algorithms** to select features . **A search strategy is processed** through the space of possible features (each subset is evaluated based on the quality of the performance of a given algorithm)

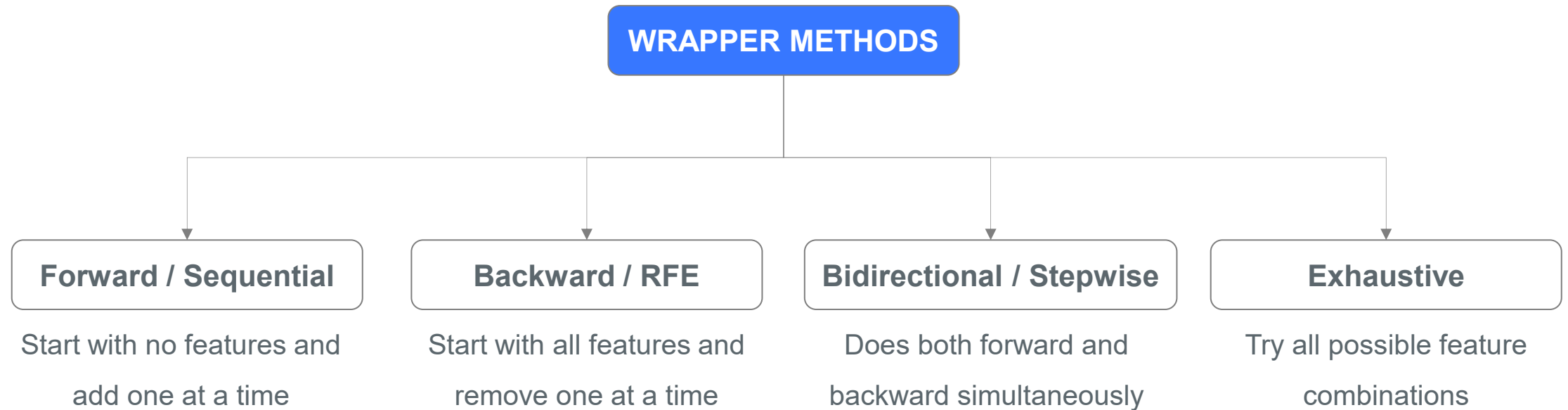
2.1. Wrapper Methods

The process



2.2. Wrapper Methods

Search Methods



2.2.1. Wrapper Methods

Search Methods – Forward / Sequential

Step 1 – Evaluate all features individually.

Step 2 - Select the one that results in the best performance

Step 3 – Test all possible combinations of the selected feature with the remaining features

Step 4 – Keep the combination that produces the best performance

Step 5 – Continue the loop by adding one feature at a time in each iteration until the pre-set criterion is achieved

2.2.2. Wrapper Methods

Search Methods – Backward / RFE (Recursive Feature Elimination)

Step 1 – Start with all the features in the dataset

Step 2 – Evaluate the performance of the algorithm

Step 3 – Remove one feature at a time and evaluate the performance of the model

Step 4 – Remove permanently the least significant feature among the remaining available ones

Step 5 – Continue the loop by removing one feature at a time in each iteration until the pre-set criterion is achieved

The difference between Backward Selection and RFE

Backward Selection - Focuses on **statistical significance (p-values)**.

"Is this variable statistically useful in explaining the target?"

RFE - Focuses on **predictive contribution to the model (model importance)**.

"Does this feature improve prediction performance?"

2.2.3. Wrapper Methods

Search Methods - Stepwise

Step 1 – Start with no features and evaluate all features individually

Step 2 – Select the one that results in the best performance

Step 3 – Test all possible combinations of the selected feature(s) with the remaining features

Step 4 – Keep the combination that produces the best performance

Step 5 – Check if the removal of one of the variables increase the performance of the model

a) If a non-significant variable is found, remove it from the model

b) otherwise, go to step 3

2.2.4. Wrapper Methods

Search Methods - Exhaustive

Step 1 – Create all possible feature combinations

Step 2 – For each subset build a model

Step 3 - Select the subset with the best performance

2.3. Wrapper Methods

Advantages

- Wrapper methods take into consideration the interaction of features
- Usually result in better predictive accuracy than filter methods

03. Embedded Methods

Embedded methods are techniques used to **select features during the model training.**

3. Embedded Methods

Introduction

Embedded methods are techniques used to select features during the model training.

3. Embedded Methods

Introduction

- Learn which features best contribute to the accuracy of the model **while the model is being created**
- Most common type are regularization methods, also called penalization methods
 - Introduce additional constraints into the optimization of a predictive algorithm
 - Bias the model toward lower complexity (fewer coefficients)
 - *Examples:* Lasso, ElasticNet and Ridge Regression
- Other algorithms allow us to obtain the feature importance
 - Feature importance define which variables are more important to make accurate predictions on the target
 - *Examples:* Decision Trees and Random Forests

3.1. Embedded Methods

Advantages

- Embedded methods take into consideration the interaction of features like wrapper methods
- Faster than wrapper methods
- More accurate than filter methods
- They find the “optimal” feature subset for the algorithm being trained
- Less prone to overfitting

Filter, Wrapper and Embedded Methods

The risk of overfitting

Filter methods

- Filter methods **do not train a predictive model** to choose features.
 - They only measure **simple statistical relationships** between each feature and the target.
- The feature selection step is **independent of the final model**.
- Because the model **was not repeatedly optimized during selection**, the algorithm is **less likely to adapt to noise** in the training data.

Filter, Wrapper and Embedded Methods

The risk of overfitting

Wrapper methods

- Wrapper methods repeatedly **train models to evaluate feature subsets**.
 - This is where overfitting risk becomes large.
- Wrapper methods try many feature combinations and choose the subset that gives the **best performance on training data (or validation data)**. If you try enough combinations, some subset will **fit random noise extremely well**.
- If done correctly with **cross-validation**, they can still perform well.
 - But they remain **the most prone to overfitting** among the three methods.

Filter, Wrapper and Embedded Methods

The risk of overfitting

Embedded methods

- Embedded methods perform feature selection **during model training**.
- These algorithms include **regularization or internal rules** that limit complexity.
- Embedded methods still **optimize using the training data**.
 - If regularization strength is not chosen properly, the model may still overfit.

Filter, Wrapper and Embedded Methods

Comparison and RECAP

Aspect	Filter	Wrapper	Embedded
Uses ML model	No	Yes	Yes
Computational cost	Low	Very high	Medium
Considers feature interactions	No	Yes	Yes
Risk of overfitting	Low	High	Medium
Speed	Fast	Slow	Moderate
Example	Correlation, Chi-square	RFE, Forward Selection	LASSO, Decision Trees

So many techniques and the results can
differ among themselves...

How to choose the variables to keep?

One possible way to identify the best features to keep:

- Apply different techniques and use the ones who are more frequent

Example

Variable	<i>MIC</i>	<i>Stepwise</i>	<i>Lasso</i>	<i>Random Forest</i>	<i>Count Yes</i>	
Variable A	Yes	Yes	Yes	Yes	4	Include
Variable B	Yes	Yes	Yes	Yes	4	Include
Variable C	Yes	Yes	No	Yes	3	Check if inclusion increase performance
Variable D	No	Yes	Yes	Yes	3	Check if inclusion increase performance
Variable E	Yes	Yes	No	No	2	Do not include
Variable F	No	No	Yes	No	1	Do not include
Variable G	No	No	No	No	0	Do not include

Obrigada!

Thank you!

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